MTH 451/551
Lecture 6

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1 Functional Iteration and Fixed Points

[Textbook reference: §2.3]

Many of the methods we have seen (and ones we will see later in this course) generate sequences x_n that converge to some limit r. The convergence of these methods can often be studied using a concept called *functional iteration*. Let I be some subset of $\mathbb{R}$, and let $g : I \to I$. Starting from some initial x_0, the **functional iterates** of g are given by

$$x_{n+1} = g(x_n).$$

Example 1. Let $g(x) = \cos(x)$. We consider the functional iteration starting from some different choices of x_0. Note that for each of the starting values, the iteration appears to converge to the same value, $r \approx 0.739085$, which is a solution to $g(x) = x$. ♦

The fact that the iteration in the example above converged to a solution to $g(x) = x$ is not a coincidence, as the following theorem tells us.

Theorem 1 (Limits of functional iteration; Textbook Theorem 2.19)**.** *Suppose that x_0 is given and that the sequence x_n is generated by functional iteration using the function g, $x_{n+1} = g(x_n)$ for $n \geq 0$. If $x_n \to r$ and g is continuous at r, then r is a **fixed point** of g, i.e. $g(r) = r$.*

Given this theorem, it is important to understand when a function g has a fixed point, and under what conditions will the functional iteration $x_{n+1} = g(x_n)$ converge to that fixed point. One useful result on the existence of a fixed point is the following.

Theorem 2 (A fixed point theorem; Textbook Theorem 2.3)**.** *If $g \in C[a, b]$ and $g([a, b]) \subseteq [a, b]$, then g has a fixed point $r \in [a, b]$.*

Proof. Let $f(x) = g(x) - x$. Since $g(a), g(b) \in [a, b]$, we have $f(a) \geq 0$ and $f(b) \leq 0$, so $f(a)f(b) \leq 0$. By the convergence of the bisection method (Textbook Theorem 2.2), f has a root $r \in [a, b]$, i.e. $g(r) = r$. □

This theorem tells us that a fixed point exists, but not whether the functional iteration converges to it. The main tool that we will use to answer that question is the **Contractive Mapping Theorem**. First, we need some definitions.

Definition 1 (Contraction; Textbook Definition 2.20)**.** *We say that g is a **contraction** (or **contractive mapping**) on $I \subseteq \mathbb{R}$ if there is a constant $0 \leq \gamma < 1$ such that*

$$|g(x) - g(y)| \leq \gamma |x - y| \qquad \text{for all } x, y \in I.$$

Lemma 1 (Textbook equation (2.12))**.** *Let I be an interval, and let $g \in C^1(I)$. Then*

$$|g(x) - g(y)| \leq \gamma |x - y| \quad \text{for all } x, y \in I \qquad \text{if and only if} \qquad |g'(x)| \leq \gamma \quad \text{for all } x \in I.$$

Proof. First, suppose that $|g'(x)| \leq \gamma$ for all $x \in I$, and let $x, y \in I$. By the Mean Value Theorem,

$$|g(x) - g(y)| = |g'(\xi)||x - y| \leq \gamma |x - y|$$

for some ξ between x and y, so $|g'(\xi)| \leq \gamma$.

Conversely, suppose that $|g(x) - g(y)| \leq \gamma |x - y|$ for all $x, y \in I$, and let $x \in I$. For any $y \in I$ with $y \neq x$,

$$\left| \frac{g(y) - g(x)}{y - x} \right| \leq \gamma.$$

Taking the limit as $y \to x$ (with $y \in I$), we obtain $|g'(x)| \leq \gamma$. □

Definition 2 (Cauchy sequence; Textbook Definition 2.21)**.** *A sequence $\{x_n\}$ in $\mathbb{R}$ is a **Cauchy sequence** if, for each $\varepsilon > 0$, there is a natural number $N = N(\varepsilon)$ such that $|x_m - x_n| \leq \varepsilon$ whenever $m, n \geq N$.*

Theorem 3 (Convergence of Cauchy sequences; Textbook Theorem 2.22)**.** *Cauchy sequences in $\mathbb{R}$ converge. In other words, if $\{x_n\}$ is a Cauchy sequence, then there is some $r \in \mathbb{R}$ such that $x_n \to r$ as $n \to \infty$.*

Theorem 4 (Contractive mapping theorem; Textbook Theorem 2.23)**.** *Let $I \subseteq \mathbb{R}$ be a closed set. If $g(I) \subseteq I$ and g is a contraction on I with contraction constant $0 \leq \gamma < 1$, then the following hold:*

1. *g has a unique fixed point $r \in I$.*
2. *If $x_0 \in I$ and $x_{n+1} = g(x_n)$, then $x_n \to r$.*
3. *The convergence is at least linear, with the error bounds*

$$|x_n - r| \leq \gamma^n |x_0 - r|, \qquad |x_n - r| \leq \frac{\gamma^n}{1-\gamma}|x_1 - x_0|.$$

Proof. We first establish uniqueness: there is **at most one** fixed point in I. To do this, we suppose that $r, s \in I$ are fixed points; we will show that r must equal s. Since they are fixed points,

$$|r - s| = |g(r) - g(s)| \leq \gamma |r - s|.$$

If $|r-s| \neq 0$, then, since $\gamma < 1$, we would have $|r-s| < |r-s|$, a contradiction. Therefore, $|r-s| = 0$, and $r = s$, establishing uniqueness. We now move on to proving existence.

Let $x_{n+1} = g(x_n)$. Since $|x_{k+1} - x_k| = |g(x_k) - g(x_{k-1})| \leq \gamma |x_k - x_{k-1}|$, using induction we see that $|x_{k+1} - x_k| \leq \gamma^k |x_1 - x_0|$ for all k. Suppose that $m > n$, and consider $|x_m - x_n|$. We have

$$\begin{aligned} |x_m - x_n| &= \left| \sum_{k=n}^{m-1} (x_{k+1} - x_k) \right| \leq \sum_{k=n}^{m-1} |x_{k+1} - x_k| \leq \sum_{k=n}^{m-1} \gamma^k |x_1 - x_0| \\ &= |x_1 - x_0| \left(\sum_{k=0}^{m-1} \gamma^k - \sum_{k=0}^{n-1} \gamma^k \right) = \frac{\gamma^n - \gamma^m}{1-\gamma} |x_1 - x_0| \leq \frac{\gamma^n}{1-\gamma} |x_1 - x_0|. \end{aligned}$$

Since the final term vanishes as $n \to \infty$, we see that $\{x_n\}$ is a Cauchy sequence, which therefore must converge to some number $r \in \mathbb{R}$. Since the terms of the sequence are all in I and I is closed, it holds that $r \in I$. Therefore, $x_n \to r \in I$. Since g is continuous at r, $g(x_n) \to g(r)$. Therefore,

$$r = \lim_{n \to \infty} x_{n+1} = \lim_{n \to \infty} g(x_n) = g(r),$$

and so r is the unique fixed point of g in I.

We proved above that the inequality

$$|x_m - x_n| \leq \frac{\gamma^n}{1-\gamma} |x_1 - x_0|$$

holds for all $m > n$, so

$$\lim_{m \to \infty} |x_m - x_n| = |r - x_n| \leq \frac{\gamma^n}{1-\gamma} |x_1 - x_0|,$$

which establishes one of the error bounds. The other error bound follows readily by induction, using $|x_{n+1} - r| = |g(x_n) - g(r)| \leq \gamma |x_n - r|$. □

Example 2. We return to Example 1, and prove that for any $x_0 \in \mathbb{R}$, the functional iterates of $g(x) = \cos(x)$ converge to the same limit r, which must be a fixed point $g(r) = r$. (Note that $x = \cos(x)$ has only one solution, so the fixed point is unique).

Let $I = [0, 1]$. Since cos is decreasing on $[0, 1]$, we have $g(I) = [\cos(1), \cos(0)] = [\cos(1), 1] \subseteq I$. For $x, y \in I$, the Mean Value Theorem gives

$$\frac{g(x) - g(y)}{x - y} = \cos'(\xi) = -\sin(\xi)$$

for some ξ between x and y, and so

$$|g(x) - g(y)| = |\sin(\xi)||x - y| \leq \sin(1)|x - y|.$$

Therefore, g is a contraction on I with $\gamma = \sin(1) \approx 0.8415$. By the contractive mapping theorem (Theorem 4), g has a unique fixed point $r \in [0, 1]$, and the iteration converges to r whenever $x_0 \in [0, 1]$.

Now let $x_0 \in \mathbb{R}$ be arbitrary. Then $x_1 = \cos(x_0) \in [-1, 1]$, and since cos is even, $x_2 = \cos(x_1) = \cos(|x_1|) \in [\cos(1), 1] \subseteq I$. Therefore, the iterates $x_2, x_3, \ldots$ are the functional iterates of g starting from $x_2 \in I$, and so $x_n \to r$. Furthermore,

$$|x_n - r| \leq \gamma^{n-2}|x_2 - r| \leq \big(\sin(1)\big)^{n-2} \qquad \text{for } n \geq 2. \qquad \blacklozenge$$

The contractive mapping theorem guarantees convergence that is *at least* linear. The following theorem gives the precise order of convergence in terms of the derivatives of g at the fixed point.

Theorem 5 (Convergence rates; Textbook Theorem 2.26)**.** *Let $\{x_n\}$ be a sequence generated by functional iteration with g, and let r be a fixed point of g, with $x_n \neq r$ for all n.*

1. *If $g \in C^1$ near r and $0 < |g'(r)| < 1$, then $x_n \to r$ precisely linearly if x_0 is sufficiently close to r.*
2. *Let $m \geq 2$. If $g \in C^m$ near r and $g^{(j)}(r) = 0$ for $1 \leq j \leq m - 1$, but $g^{(m)}(r) \neq 0$, then $x_n \to r$ with precisely order m if x_0 is sufficiently close to r.*

Proof. In either case, there is a constant $\gamma \in [0, 1)$ and an interval $[r - \delta, r + \delta]$ with $\delta > 0$ on which $|g'(x)| \leq \gamma$. By Lemma 1, g is a contraction on $[r-\delta, r+\delta]$. Furthermore, for $x \in [r-\delta, r+\delta]$, taking $y = r$ and using $g(r) = r$, we have $|g(x) - r| \leq \gamma|x - r| \leq \delta$, so $g(x) \in [r - \delta, r + \delta]$. By the contractive mapping theorem (Theorem 4), convergence is guaranteed if $x_0 \in [r - \delta, r + \delta]$.

In the first case,

$$\frac{|x_{n+1} - r|}{|x_n - r|} = \frac{|g(x_n) - g(r)|}{|x_n - r|} = |g'(\xi_n)|$$

for some ξ_n between x_n and r. Therefore, $|x_{n+1} - r|/|x_n - r| \to |g'(r)| \in (0, 1)$, and we see that the rate of convergence is precisely linear. In the second case, repeated applications of l'Hôpital's rule reveal that

$$\lim_{x \to r} \frac{g(x) - r}{(x - r)^m} = \lim_{x \to r} \frac{g'(x)}{m(x - r)^{m-1}} = \cdots = \lim_{x \to r} \frac{g^{(m)}(x)}{m!} = \frac{g^{(m)}(r)}{m!},$$

so $|x_{n+1} - r|/|x_n - r|^m \to |g^{(m)}(r)|/m! \neq 0$, and the rate of convergence is precisely order m. □

Example 3. We return again to Example 1, and show that the convergence is precisely linear. We have $g'(r) = -\sin(r)$. Since $r \in (0, 1)$, we have $0 < \sin(r) < \sin(1) < 1$, so $0 < |g'(r)| < 1$ (numerically, $|g'(r)| = \sin(r) = \sqrt{1 - r^2} \approx 0.6736$). By Theorem 5, the convergence is precisely linear if x_0 is sufficiently close to r.

Since we showed above that $x_n \to r$ for *any* x_0, the iterates eventually come arbitrarily close to r, and so the convergence is precisely linear for any x_0 (provided $x_n \neq r$ for all n). To see this directly, by the Mean Value Theorem,

$$\frac{|x_{n+1} - r|}{|x_n - r|} = \frac{|g(x_n) - g(r)|}{|x_n - r|} = |\sin(\xi_n)|$$

for some ξ_n between x_n and r. Since $x_n \to r$, also $\xi_n \to r$, and so

$$\frac{|x_{n+1} - r|}{|x_n - r|} \to \sin(r) \approx 0.6736.$$ ♦